CHINMAYI R M

chinmayirm.04@gmail.com | LinkedIn | GitHub

SKILLS

Key Courses: Big Data, Mathematics for Machine Learning, Statistics for Data Science, Advanced Data Analytics, Social Computing, Database Management Systems, LLM
Skills: Python, Data Analysis, Data Science, Natural Language Processing (NLP), HTML/CSS, NumPy, Pandas, Hadoop, Apache Spark, Apache Kafka, Tensorflow, Computer Vision, Machine Learning, Big Data, R, SQL, Apache Hive, C/C++, Data Structures & Algorithms, Deep Learning, Pytorch

EDUCATION

PES University **September 2022 - May 2026**
Bachelor's, Computer Science *GPA: 8.97*
- Bachelor of Technology in Computer Science and Engineering (AI & ML)
- Activities: PES MUN Society - Co-Editor, FTR editorial | TEDxPESU - Curator | Apple Developers Group - AIML Domain - Member | Student Interest Group on “Time Series and Data Analytics” at the Centre for Data Science and Applied Machine Learning (CDSAML), PES University

PROFESSIONAL EXPERIENCE

Indian Space Research Organisation (ISRO) - RRSC South **June 2025 - July 2025**
Summer Intern *Bengaluru, KA, India*
- Built an end-to-end pipeline for remote sensing–based agricultural monitoring using deep learning.
- Developed a deep learning model to predict crop distribution using 10m-resolution, 13-band satellite imagery a curated spectral library.
- Utilised spatio-temporal satellite data to capture seasonal and geographic crop variation.
- Built an interactive Streamlit dashboard to visualize crop predictions and satellite inputs dynamically.
- abc

Tropes and Pivots Learning Systems Pvt. Ltd. **May 2024 - August 2024**
Machine Learning Intern
- Implemented a scalable engine to retrieve and rank courses from an SQL database, leveraging NLP for optimised query interpretation.
- Created a dynamic recommendation engine that provides curated educational content based on user queries, integrating machine learning algorithms and web scraping techniques to enhance content personalisation

Centre for Data Science and Applied Machine Learning, PES University **June 2024 - July 2024**
Research Intern
- Introduced quantum computing elements in the multihead self-attention and feedforward components of the Vision Transformer architecture, enhancing the model’s ability to capture complex patterns.
- Utilized IBM's Qiskit to implement Quantum Vision Transformers, developing a custom decoder specifically for object detection tasks.
- Collaborated with an industry mentor, contributing to a research paper currently in process.

PES University **January 2024 - May 2024**
Teaching Assistant
- TA for course UE22AM241B : Mathematics for Machine Learning

KEY PROJECTS

Yoda Bot | Python, NLP, Sentiment Analysis, Hugging Face Transformers, DialoGPT-small, PyTorch, Streamlit
- Designed an AI-powered chatbot to emulate Yoda from the popular Star Wars movie franchise, offering wisdom-laced responses aligned with user emotions.
- Integrated a pre-trained sentiment analysis model to tailor emotionally aware responses.
- Fine-tuned DialoGPT-small on Yoda’s dialogues extracted from the Cornell Movie Dialogs Corpus to mimic the character's speech.
- *Link to project*

Chroma Vision | Python, NumPy, Tensorflow, OpenCV, Scikit-learn, Matplotlib

- Implemented colour detection and correction for colour blind individuals (Protanopia)
- Developed a CNN model using TensorFlow and Keras for color correction.
- Trained the model on CIFAR-10 dataset.
- Utilized OpenCV and K-Means clustering for colour detection.
- *Link to project*

Compression Analysis Using Machine Learning | Python, cv2, NumPy, Pandas
- Compared Lossy, Principal Component Analysis and Singular Value Decomposition compression
- techniques.
- Used Machine Learning to evaluate methods based on PSNR and compression ratio.
- Evaluation performed on diverse image set to ensure robustness.
- *Link to project*

Data Analysis of Airbnb Listings | R programming, R studio
- Performed Exploratory Data Analysis (EDA) on the dataset to understand the underlying
- patterns and relationships.
- Implemented data cleaning techniques to handle missing values and removed outliers to ensure
- the accuracy of the analysis.
- Developed functions to extract specific insights from the data.
- *Link to project*

RESEARCH EXPERIENCE

Custom Augmented Model for Large-scale Latin Language Identification
- Utilized the strengths of **transformer architecture** and made **custom optimizations** in **extracting token embeddings, pooling mechanisms, and the classification head** to enhance performance in Latin language identification.
- Conducted iterative adjustments to two custom augmented models, focusing on optimizing transformer components, data preprocessing, and model architecture.
- Conducted comprehensive **A/B testing** to compare the performance of both models, evaluating key metrics like accuracy, precision, recall, and F1-score.
- Applied rigorous **hypothesis testing and statistical analysis** to validate the impact of each tweak, identifying the **most effective optimizations** for future language classification tasks.

Quantum Vision Transformer for Object Detection
- Introduced **quantum computing** elements into the multihead self-attention and feedforward components of the **Vision Transformer architecture**, significantly enhancing the model's capacity to capture **complex patterns.**
- Implemented Quantum Vision Transformers using **IBM's Qiskit,** including the development of a **custom decoder** tailored for **object detection tasks.**
- Collaborated with an **industry mentor** on a research paper in progress, focusing on advancements in quantum-enhanced machine learning models.

Geopolitical Advisory System
- Developed a context-aware system using **Retrieval-Augmented Generation (RAG)** and **LlamaIndex** to analyse **UN Security Council resolutions, socioeconomic data, and voting patterns.**
- The system identifies **patterns, contradictions, and correlations** while visualising insights through **social computing** and **graph-based techniques.**
- The system delivers **precise insights and predictions**, simplifying the analysis of global policies.

ACHIEVEMENTS

Prof. C N Rao Scholarship x 4
- Awarded to top 20% students in each batch. Received for 4 consecutive semesters.

Diplôme d'études en langue française - B2
- Issued by France Éducation international .

Junior CTO Valedictorian
- Issued by Dr. Lina Fruzzetti, faculty, Brown University.
- Was also selected as a finalist for the Innovation Summit held by Indian Institute of Management (IIM), Rohtak.